Residue Orientation and Root Transfer

Smaller explicit Diophantine equations for syntactic consistency of ZFC

AI-assisted research prepared for Vladimir Reshetnikov
A contribution for the ProveIt repository

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Abstract

We give two proof-driven reductions of explicit Diophantine consistency equations. First, a signed Pell argument recovers a coordinate as $\pm 2^R$ with R odd; affine congruences then select the positive sign. This removes a dedicated square witness and gives a quartic in forty unrestricted integer variables. Exact coordinate eliminations give a sextic in thirty-four and an octic in twenty-nine integer variables. Second, we transfer Pell roots into a transport gap. A two-root norm gives twelve strictly positive variables of degree eighty-four in 103 arithmetic operations. A three-root extension uses a coprimality argument to recover an auxiliary integer root, giving eleven positive variables of degree 325 in 113 operations. A thirteen-positive-variable companion has degree thirty in 94 operations. These improve specified points in the inspected repository snapshot. Every fixed coefficient is supplied exactly, with an offline verification recipe. The new equivalences are proved by ordinary mathematics using explicitly named inherited compiler, Pell, and syntactic calibration theorems. They are not new end-to-end proof-assistant certificates, and no global minimality or proof of consistency is asserted.

Source snapshot. `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`
Primary deliverables.
`integer/quartic40_direct.json`
`eleven/p11_positive_degree325.json`
`positive/p12_positive_degree84.json`
Reproduction. `python3 verify.py`; `python3 verify.py -optimized`.

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1 Results, comparison, and the meaning of simplicity

Fix the repository's recursive syntactic proof predicate for ZFC. In this article Con(ZFC) means that no finite derivation of falsity exists for that predicate. The target is an explicitly specified polynomial $P \in \mathbb{Z}[x_1, \ldots, x_m]$, together with a stated domain D, for which ordinary mathematics formalizable in ZFC proves

$$(\exists \mathbf{x} \in D^m \; P(\mathbf{x}) = 0) \quad \Longleftrightarrow \quad \text{there is a finite syntactic ZFC refutation.} \tag{1}$$

Negating the two sides yields exactly the requested equivalence between unsolvability and Con(ZFC). This establishes a relationship between two statements; it does not establish their truth or falsity.

The phrase "simplest equation" has several competing meanings. A reduction in variables can increase degree, and treating a large fixed integer as one arithmetic-circuit leaf can hide most of the written description. We therefore report witness domain and count, exact total degree, number and degree of residuals, arithmetic operations when optimized, and fixed coefficient storage separately. A named intermediate register is a polynomial abbreviation, never an additional quantified unknown.

Equation	Witness domain	Variables	Degree	Binary operations
Q_{40}^{dir}	$\mathbb{Z}$	40	4	Not optimized
Q_{34}	$\mathbb{Z}$	34	6	Not optimized
Q_{29}	$\mathbb{Z}$	29	8	Not optimized
P_{11}	$\mathbb{Z}_{>0}$	11	325	113
P_{12}	$\mathbb{Z}_{>0}$	12	84	103
P_{13}	$\mathbb{Z}_{>0}$	13	30	94

Table 1: New single polynomial equations. The integer equations are sums of 25, 19, and 14 squared residuals, respectively.

Theorem 1.1 (The delivered consistency equations)**.** *Each polynomial in Table 1, with its own complete fixed coefficient tuple and stated witness domain, satisfies* (1)*. Its exact degree and number of variables are as listed. The claims use the inherited source theorems identified in the proofs and provenance files; they do not introduce a consistency assumption.*

The integer constructions improve the inspected degree-four, degree-six, and degree-eight counts from $41, 35, 30$ to $40, 34, 29$. The parent quartic is package 64's Q_{41}; the two smaller-count comparison companions are package 62's sextic and octic. The twelve-positive construction improves package 61's degree-118, 104-operation equation in both measures: $118 \to 84$ is a 28.81% degree reduction, and $104 \to 103$ saves one operation. The thirteen-positive comparison is a tradeoff: degree $34 \to 30$, with operations $93 \to 94$. These are comparisons to literal artifacts in the pinned repository, not just to the collection README, which still described an earlier wave at the inspection time. Relevant comparison files and their hashes are retained in `baseline/` and `BASELINE.json`.

The eleven-positive extension improves package 52's degree-954, 113-operation point to degree 325 at the same 113-operation cost. This is a 65.93% degree reduction. Its two residuals have degrees 312 and 13. A modulo-four unit-product argument combines them without taking their sum of squares; the final polynomial has degree $312 + 13 = 325$.

The small-variable high-degree records remain separate: the repository already contains nine-positive and ten-unrestricted-integer constructions. The present work does not lower those counts. Likewise, the fifteen-positive degree-24 and 66-operation constructions address different objectives.

Slice	Inspected predecessor	New result	Nature of the change
Quartic over $\mathbb{Z}$	41 variables, package 64	40	One sign witness removed
Sextic over $\mathbb{Z}$	35 variables, package 62	34	Same orientation, exact elimination
Octic over $\mathbb{Z}$	30 variables, package 62	29	Same orientation, exact elimination
Eleven positive	Degree 954 / 113 ops, package 52	325 / 113	Three roots in one transport carrier
Twelve positive	Degree 118 / 104 ops, package 61	84 / 103	A strong root is eliminated
Thirteen positive	Degree 34 / 93 ops, package 61	30 / 94	Adaptive modulus composition

Table 2: A bounded comparison, with domains and costs kept distinct.

The recommended Q_{40}^{dir} has 393 expanded symbolic monomials, compared with 388 in its parent; the alternate projection parent has 614. No coefficient-height, expanded-length, or global circuit-cost reduction is claimed for that family. A mathematically new mechanism can improve one boundary of the comparison without dominating every other equation.

1.1 What is new in this iteration

The first method changes the order in which signs are established. A positive-square guard is usually imposed before a Pell argument. Here the argument is proved with signed X, Y, k, c throughout, and gives $X = \pm 2^R$ with R odd. The already imposed congruence $X \equiv 2 \pmod 3$ then rules out the negative branch. The congruence is enforced most directly by $X_{\mathrm{old}} = 3X_3 - 1$, so it needs no new witness and preserves quadratic residual degree. A companion version restricts the two existing scale coordinates instead, retaining a monic X definition for the sextic and octic projections.

The second method exploits an existing free positive transport gap. Instead of keeping a strong root f and a gap r separately, use the coordinate $n = r - L + f$, with $L > |f|$, and eliminate f by a norm. The proof must handle both signed roots recovered by that norm. A stronger transport multiplier guarantees the inverse map has $n > 0$. The remaining auxiliary relations use only f^2, so their degree remains controlled after substitution.

The eleven-variable extension uses that normalization to make the three radicands independent of the signs of their roots. It retains the adaptive main Pell equation separately and puts the first, strong, and auxiliary roots into the transport carrier. Multiplying the normalized strong ordinate by $q(q-1)$ makes the auxiliary denominator coprime to the transport denominator, so a rational root recovered from the norm must be an integer. Elementary separation bounds exclude every irrational cancellation. Norms, positive majorants, and unit-product finalizers already appear in earlier packages; the new claim is this particular arithmetic composition and its measured endpoint.

Both mechanisms are new relative to the inspected source packages. We make no literature-wide priority claim. The thirteen-variable adaptive carrier and the low-degree coordinate eliminations explicitly reuse earlier ideas; their attribution is retained in the component notices.

1.2 Why the final constructions retain the native endpoint

The user permitted a fresh computational or logical source. We examined direct provability encodings as well. Carl and Moroz give an explicit Diophantine treatment of provability in predicate calculus and discuss set-theoretic applications [4]. Formisano, Gavazzi and Omodeo investigate a relational equational presentation of NBG; their 2025 article describes preliminary work on its proper-axiom component [5]. These are useful routes to smaller transparent proof checkers, but our inspection did not produce a fully materialized alternative with a competitive, proved coefficient and variable count. An uncompiled proof checker would not be a replacement for the requested explicit equation.

The delivered constructions therefore retain the existing native K215-to-polynomial source and improve its arithmetic representation. They do not use Jones's universal polynomial or Sun's ten-integer finalizer. Jones's work [3] and Sun's recent ten-variable theorem [6] explain relevant comparison boundaries, but are not newly invoked to justify these polynomial equations. Native compiler semantics and K215 calibration remain explicit inherited dependencies.

2 Residue orientation: a quartic in forty unrestricted integers

The main structural improvement of this contribution is the removal of a dedicated positivity witness. Source 64 uses the equation $X = 2u^2$ to start its Pell argument with $X \geq 0$. We allow X to have either sign, recover $X = \pm 2^R$ with R odd, and select the positive branch by the congruence $X \equiv 2 \pmod 3$. The congruence is obtained by affine restrictions on two existing coordinates, so it introduces no unknown. The preferred quartic uses the even simpler direct restriction $X = 3X_3 - 1$ and has 393 expanded symbolic monomials. We first prove the equivalent 40-variable version with two affine restrictions, whose monic X definition supports the best sextic and octic projections; the direct version is specified in Subsection 2.6.

Theorem 2.1. *There is an explicitly specified fixed polynomial Q_{40}^{aff} in forty unrestricted integer unknowns, of exact total degree four, such that ordinary mathematics formalizable in ZFC proves*

$$(\exists \mathbf{x} \in \mathbb{Z}^{40})\ Q_{40}^{\mathrm{aff}}(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{there is a finite syntactic ZFC refutation.}$$

Consequently ZFC proves that the absence of integer zeros of Q_{40}^{aff} is equivalent to Con(ZFC). *Exact polynomial projections give corresponding equations in thirty-four integers of degree six and twenty-nine integers of degree eight.*

The statement imports the explicitly identified native-compiler, fixed-tuple, K215 calibration, and Pell-rank theorems of the repository. The new part is the sign-uniform power-recovery argument below and its application to literal polynomials. It is an ordinary proof, with exact finite algebraic checks and an independent review; it is not a new end-to-end Lean/Rocq certificate or an exported derivation in an internally encoded ZFC calculus.

2.1 The fixed integers and the complete quadratic system

Use the whole source34 `mixed-parity-unrounded-1064921` tuple. Its ports are

$$\beta = 2^{1064921} - 1, \qquad K_0, \qquad M_C, \qquad M_F.$$

The last port is the already shifted high mask. All four literal integers are supplied as lossless gzip payloads of minimal unsigned little-endian bytes. Their bit lengths are respectively

1064921, 1065043, 1064921, 1064922; their combined uncompressed payload is 532479 bytes. The included standard-library verifier reconstructs every byte from the retained 943-tile list and checks the original hashes. No coefficient is selected by assuming consistency. The relevant fixed facts are

$$\beta \geq 17, \qquad 0 < M_C < \beta < M_F < 2\beta, \qquad K_0 \geq 0, \qquad \beta \equiv 1 \pmod 3. \tag{2}$$

The thirty-one core unknowns are

$$J_3, F, \alpha, z, f, h, r, v, w_3, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_0, \Delta.$$

There are also three private triples $\mathbf{u}, \mathbf{v}, \mathbf{e}$, for a total of forty unknowns. In the JSON packet, v is `v_scale`, B_0 is `Baux`, and the triples are `FA_pos1,...,3`, `FC_pos1,...,3`, and `eta_zeta_pos1,...,3`. Every coordinate ranges over all integers. Define only the following polynomial abbreviations:

$$J_0 = 3J_3 + 1, \qquad W = 4(3w_3 + 1), \qquad q = \beta J_0 + 1,$$
$$C = q - F - \alpha, \qquad k = \eta + \zeta,$$
$$\mathcal{R} = (q - F)(N - q) - CL + (M_C + qM_F)J_0.$$

Also set $\mathrm{Pos}(x_1, x_2, x_3) = 1 + x_1^2 + x_2^2 + x_3(x_3 + 1)$. The exact main equation is $Q_{40}^{\mathrm{aff}} = \sum_{\nu=1}^{25} r_\nu^2 = 0$, where the twenty-five residuals are the following quadratic polynomials:

$$r_1 = L - q^2 + 1, \qquad r_2 = N - q(L + 1),$$
$$r_3 = X - qW, \qquad r_4 = a - E - (2v + 1)N,$$
$$r_5 = E(X + 1) - Xa, \qquad r_6 = c - \eta + (E - a)k,$$
$$r_7 = P - k - E(c - \eta), \qquad r_8 = \tau^2 - P^2 + Pk - 1,$$
$$r_9 = D - X - ac - \gamma(4a + 3), \qquad r_{10} = \Delta - a^2 - 4a - 3,$$
$$r_{11} = C_2 - c^2, \qquad r_{12} = D^2 - C_2\Delta - 1.$$

$$r_{13} = k - 1 - hE - \mathcal{R}, \qquad r_{14} = (K_0 + W)C - z(q - 1) - F,$$
$$r_{15} = F_2 - f^2, \qquad r_{16} = V + c - fo,$$
$$r_{17} = V - jc + k - 1 - hE, \qquad r_{18} = V^2 + g - y^2,$$
$$r_{19} = I - (2r + 1)C_2, \qquad r_{20} = B_0 - I^2,$$
$$r_{21} = B_0 - \Delta(F_2 - 1), \qquad r_{22} = y^2 - 1 - B_0 g,$$
$$r_{23} = F\alpha - \mathrm{Pos}(\mathbf{u}), \qquad r_{24} = FC - \mathrm{Pos}(\mathbf{v}),$$
$$r_{25} = \eta\zeta - \mathrm{Pos}(\mathbf{e}).$$

No abbreviation adds an existential unknown. Expanding these rows gives exactly the literal packet `quartic40.json`. Each row has degree two. Only r_8 contains τ, and its coefficient of τ^2 is one; therefore $[\tau^4]Q_{40}^{\mathrm{aff}} = 1$, proving exact degree four on the actual fixed-coefficient slice.

2.2 What changed from the parent

In source64's 41-variable quartic, make the affine substitutions

$$J_{\mathrm{old}} = 3J_3 + 1, \qquad w_{\mathrm{scale,old}} = 3w_3 + 1,$$

then delete the residual $X - 2u^2$ and its private coordinate u. The other twenty-five rows are unchanged under those substitutions. Since $\beta \equiv 1 \pmod 3$, the new rows imply

$$q \equiv 2 \pmod 3, \qquad W \equiv 1 \pmod 3, \qquad X = qW \equiv 2 \pmod 3. \tag{3}$$

The work is to prove that this residue restriction can replace the deleted positivity witness without introducing spurious zeros.

2.3 Bounds before any sign is chosen

At an integer zero every residual is zero. The form Pos is strictly positive on $\mathbb{Z}^3$ and represents every positive integer. To see surjectivity, apply Legendre's three-square theorem to $4n - 3$, which is 1 modulo 4 and cannot have the exceptional form $4^a(8b + 7)$; its representation has one odd and two even squares.

The two guards $F\alpha > 0$ and $FC > 0$ force F, α, C to have a common nonzero sign. Thus $q = F + \alpha + C$ has that sign and $|q| \geq 3$. Write $Q = |q|$. Because $q = \beta J_0 + 1$, one has $J_0 \neq 0$ and $Q \geq \beta - 1 \geq 16$. Set

$$R = ((q - 1)(q - F) + \alpha)(q^2 - 1) + (M_C + qM_F)J_0.$$

The first two rows give $L = q^2 - 1$, $N = q^3$, and hence $\mathcal{R} = R$. The sign-free packing estimate is

$$Q(Q^2 - 1) < R < Q^4 - 2. \tag{4}$$

Here is a direct reminder of its proof. If $q = Q > 0$, put $G = Q(Q - F) - C$. Then

$$Q + 2 \leq G \leq Q^2 - Q - 1, \quad 0 < (M_C + QM_F)J_0 < (2Q + 1)(Q - 1).$$

If $q = -Q < 0$, put $F_0 = -F$, $C_0 = -C$; now $G = Q(Q - F_0) + C_0$ satisfies

$$2Q + 1 \leq G \leq Q^2 - 2, \quad 0 < (QM_F - M_C)(Q + 1)/\beta < 2Q(Q + 1).$$

Multiplying the displayed bounds for G by $Q^2 - 1$ gives (4) in both cases. This uses no sign of X, Y, k.

Let $s = 2v + 1$, $i = 2r + 1$, and $Y = sq^3$. Rows r_4, r_5 give $a = E + Y$ and $E = XY$; thus $a = (X + 1)Y$. The nonzero integer $W = 12w_3 + 4$ has $|W| \geq 4$. Since s is odd, writing $U = |X|$, $Z = |Y|$, and $M = |E|$ gives

$$U \geq 4Q \geq 64, \qquad Z \geq Q^3 \geq 4096, \qquad M = UZ \geq 4Q^4 > 3R + 1. \tag{5}$$

The last guard implies that η, ζ have a common nonzero sign. Put

$$\xi = \operatorname{sgn} X, \quad \sigma = \operatorname{sgn} k, \quad \delta = \operatorname{sgn} Y, \quad K = |k|, \quad \eta_0 = |\eta|, \quad c_0 = |c|.$$

Then $K \geq 2$, $0 < \eta_0 < K$, and

$$c_0 = KZ + \delta\eta_0, \qquad K(Z - 1) < c_0 < K(Z + 1), \qquad \operatorname{sgn} c = \sigma\delta. \tag{6}$$

Set $A = a + 2$, $H = 4a + 3$, and define the positive absolute Pell parameters

$$B = |A| = (U + \xi)Z + 2\xi\delta, \qquad P_* = |2XY^2 + 1| = 2UZ^2 + \xi.$$

Their signs in these expressions are valid because $(U - 1)Z > 2$. They satisfy

$$B \geq (4Q - 1)Q^3 - 2 > Q^4 > R, \qquad P_* > B \geq 2, \qquad c_0 > K. \tag{7}$$

For instance $P_* - B \geq 2UZ^2 - (U + 1)Z - 3 > 0$ in (5). These inequalities hold in every sign branch.

2.4 A sign-uniform Pell power-recovery argument

For $b \geq 2$, use the standard Pell polynomials

$$\chi_b(t) + \psi_b(t)\sqrt{b^2-1} = (b + \sqrt{b^2-1})^t.$$

The following proof retains all signs until the last congruence.

Classify the first and main norms. The source rows imply

$$\tau^2 - XY^2(XY^2+1)k^2 = 1, \qquad D^2 - (B^2-1)c^2 = 1.$$

Consecutive-product Pell classification gives positive indices n, p with

$$|\tau| = \chi_{P_*}(n), \quad K = 2\psi_{P_*}(n), \quad |D| = \chi_B(p), \quad c_0 = \psi_B(p). \tag{8}$$

There is no hidden positive-X hypothesis: if $X < 0$, put $m = UZ^2$; the first coefficient is $m(m-1) = (m-1)m$ and its positive Pell parameter is $2m-1 = P_*$. Since $P_* > B$ and $c_0 > K$, monotonicity of ψ implies $p > n$. Otherwise $c_0 = \psi_B(p) \leq \psi_{P_*}(n) = K/2$, a contradiction.

Establish the ordinary-rank hypotheses. Modulo $M = |E|$, the parameter P_* is congruent to ξ. The recurrence for ψ gives $\psi_{P_*}(n) \equiv \xi^{n-1} n \pmod{M}$. The index row $k = R + 1 + hE$ therefore gives

$$\sigma\xi^{n-1}2n \equiv R+1 \pmod{M}. \tag{9}$$

If the sign on the left is positive, $n \geq (R+1)/2$. If it is negative, then $2n \geq M - R - 1 > 2R$, so $n > R$. Thus in both branches $p > n \geq (R+1)/2 \geq 24$. The elementary bound $\psi_B(p) \geq (2B-1)^{p-1}$ gives

$$c_0 > B(B^2-1)^2, \qquad c_0 > 2p, \qquad c_0 > 2R. \tag{10}$$

The first already follows for $p \geq 6$; the last also follows from $c_0 \geq \psi_B(2) = 2B > 2R$.

The ordinary strong equation is

$$(ic^2)^2 = (B^2-1)(f^2-1), \qquad i = 2r+1 \neq 0.$$

The inherited ordinary strong-rank lemma, now with its size hypothesis proved in every sign branch, supplies an m such that

$$|f| = \chi_B(m), \qquad p \mid m, \qquad c_0 \mid m, \qquad m > 2p. \tag{11}$$

The exact source is the retained `PELL_RELAXED_AUXILIARY_PROOF.md`; its signed use and the next step are detailed in the retained odd-scale and signed-domain notes. Only $B, c_0, |f|, |i|$ occur in its hypotheses.

Recover the main rank before orienting signs. Let $\rho = |i|c_0^2$. The auxiliary equations and exposed quotients give

$$\rho^2V^2 - (\rho^2-1)y^2 = 1, \qquad V \equiv -c \pmod{|f|}, \qquad V \equiv -R \pmod{c_0}.$$

For an odd auxiliary index $\ell = 2t+1$ and a sign e_0, Pell classification gives $V = e_0 Q_t(\rho^2)$, where

$$Q_t(1-B^2) = (-1)^t\psi_B(2t+1), \qquad Q_t(0) = (-1)^t(2t+1).$$

Here $\rho^2 \equiv 1 - B^2 \pmod{|f|}$ and $\rho^2 \equiv 0 \pmod{c_0}$. Squaring the first quotient congruence removes both e_0 and the sign of c. The inherited signed Pell step-down theorem, using $0 < 2p < m$, gives $\ell \equiv \pm p \pmod{m}$; the second congruence gives $\ell \equiv \pm R \pmod{c_0}$. Since $c_0 \mid m$, one obtains $R \equiv \pm p \pmod{c_0}$. The intervals in (10) imply

$$p = R. \tag{12}$$

This is precisely the sign-insensitive auxiliary interface used by the parent, with its prerequisites newly established also for $X < 0$.

Unwrap the first index. Since $n < p = R$, the negative-sign branch of (9), which required $n > R$, is impossible. In the positive branch write $2n = R + 1 + v_0 M$, with $v_0 \geq 0$. If $v_0 \geq 1$, then $M > 3R + 1$ again implies $n > R$. Therefore

$$\sigma \xi^{n-1} = 1, \qquad 2n = R + 1, \qquad R \text{ is odd.} \tag{13}$$

One ratio estimate handles all signs. Put $r_0 = n - 1 = (R - 1)/2 \geq 2$. The elementary upper and lower Pell growth bounds give

$$\frac{c_0}{K} = \frac{\psi_B(2r_0 + 1)}{2\psi_{P_*}(r_0 + 1)} \geq \frac{1}{2}\left(\frac{(2B-1)^2}{2P_*}\right)^{r_0}.$$

For $U \geq 64$ and $Z \geq 6$,

$$\frac{(2B-1)^2}{2P_*} > U - 3. \tag{14}$$

Indeed $B \geq (U-1)Z - 2$ and $P_* \leq 2UZ^2 + 1$, and

$$\begin{aligned}(2(U-1)Z - 5)^2 - (U-3)(4UZ^2 + 2) \\ = 4(U+1)Z^2 - 20(U-1)Z + 31 - 2U \\ = U(4Z^2 - 20Z - 2) + (4Z^2 + 20Z + 31) > 0.\end{aligned}$$

Both final parentheses are positive for $Z \geq 6$. Together with (6), this implies

$$Z + 1 > \frac{(U-3)^{r_0}}{2} \geq \frac{61^{r_0}}{2}.$$

As $61^{r_0}/2 - 1 > 2^{2r_0+1}$ for $r_0 \geq 2$, we obtain

$$Z > 2^R. \tag{15}$$

The verifier checks the exact polynomial identity and positive coefficient certificates after shifting $U = 64 + u$, $Z = 6 + z$; the displayed argument is the universal proof rather than a finite-box test.

Exclude reciprocal roots and recover a signed power. Let $e = \operatorname{sgn} c$ and write $D = \lambda \chi_A(R)$ with $\lambda \in \{1, -1\}$. Since R is odd, $\psi_A(R) = \psi_B(R)$, even for $A < 0$; hence $c = e\psi_A(R)$. The polynomial Pell congruences, valid for either sign of A, H, are

$$\chi_A(R) - a\psi_A(R) \equiv 2^R \pmod{H}, \qquad \chi_A(R) + a\psi_A(R) \equiv 2^{-R} \pmod{H}. \tag{16}$$

The odd modulus makes 2 invertible. At index one the two congruences are $A - a = 2$ and $2(A + a) \equiv 1 \pmod{H}$; the recurrence proves all indices.

If $\lambda = -e$, the row $D - ac = X + \gamma H$ gives $X2^R \equiv -e \pmod{H}$. But

$$1 < U2^R < UZ < 4(U-1)Z - 4 \le |H| - 1,$$

by (15). Neither sign of $X2^R$ can be congruent to either unit modulo $|H|$. This branch is impossible. If $\lambda = e$, the same row gives $X \equiv e2^R \pmod{H}$; now

$$|X - e2^R| \le U + 2^R < U + Z < 4(U-1)Z - 3 \le |H|.$$

This multiple of H must be zero. We have proved, before assuming any positive sign of X,

$$X = e2^R, \qquad R \text{ odd}. \tag{17}$$

2.5 Orientation, soundness, and canonical completeness

Because R is odd, $2^R \equiv 2 \pmod 3$, while $-2^R \equiv 1 \pmod 3$. Combining (17) with (3) forces $e = 1$ and $X = 2^R$. The integer

$$u = 2^{(R-1)/2}$$

therefore restores the deleted row $X = 2u^2$. Restore also $J_{\text{old}} = 3J_3 + 1$ and $w_{\text{scale,old}} = 3w_3 + 1$. Every parent residual now vanishes literally. The source64 quartic41 theorem gives the finite ZFC refutation. Its theorem is invoked only after its square guard has been restored, so no positive-X argument has been used circularly.

Conversely, a finite refutation supplies the canonical source34 witnesses used in quartic41 completeness. With $d = 1064921$, they satisfy

$$q = 2^t, \quad t \text{ odd}, \quad R \equiv 3 \pmod 4, \quad X = 2^R, \quad R - t \ge 2 \text{ even},$$
$$J_{\text{old}} = \frac{2^t - 1}{2^d - 1}, \qquad w_{\text{scale,old}} = 2^{R-t-2}.$$

Modulo 3 the first old coordinate is 1, since $\beta \equiv 1$ and $2^t \equiv 2$. The second is 1 because its exponent is nonnegative and even. Thus

$$J_3 = \frac{J_{\text{old}} - 1}{3}, \qquad w_3 = \frac{w_{\text{scale,old}} - 1}{3}$$

are integers. Retain all other coordinates and discard u to obtain a zero of Q_{40}^{aff}. It is the canonical completeness family that is preserved; no claim is made that every parent zero lifts to the two affine restrictions. This proves Theorem 2.1 for Q_{40}^{aff}.

2.6 The smaller direct-coordinate quartic

For the preferred main quartic, retain the parent J, w_{scale} and replace only X by $3X_3 - 1$, while deleting $X - 2u^2$ and u. Denote this polynomial by Q_{40}^{dir}; it is the complete packet in `quartic40_direct.json`. Equivalently, use all twenty-five rows of Subsection 2.1 with the core list modified by $J_3 \mapsto J$, $w_3 \mapsto w_{\text{scale}}$, and $X \mapsto X_3$, and now use the abbreviations

$$J_0 = J, \qquad q = \beta J + 1, \qquad W = 4w_{\text{scale}}, \qquad X = 3X_3 - 1.$$

The other abbreviations and the nine private guard coordinates are unchanged. The resulting sum of squares has 40 unknowns, 25 quadratic residuals, exact degree four, and 393 expanded symbolic monomials. Its coefficient of τ^4 remains one.

Here $X \equiv 2 \pmod 3$ gives $X \neq 0$ immediately. The retained equation $X = qW$ and $q \neq 0$ then give $W \neq 0$, so $|W| \geq 4$ and $|X| \geq 4|q|$. Thus every bound and every step of the sign-uniform argument applies without change, recovering $X = e2^R$ with R odd. The residue selects $e = 1$, after which $u = 2^{(R-1)/2}$ and $X_{\text{old}} = 3X_3 - 1$ restore the parent. For completeness, canonical $X_{\text{old}} = 2^R$ with R odd gives the integer $X_3 = (2^R + 1)/3$; keep J, w_{scale} and all remaining coordinates and discard u. This direct version needs no separate residue restriction on J, w_{scale} or on β.

The two-affine-coordinate version is retained for the projections below: there $X = Wq$ is a monic polynomial definition. Directly solving $3X_3 - 1 = Wq$ would require division by 3, which is not an allowed integer-polynomial restoration and is not used here.

2.7 The exact sextic and octic projections

Deleting a row $v - P(\text{other coordinates})$ and substituting its integer-polynomial right-hand side everywhere gives a bijection of integer zero sets. The inverse restores v. The transformations below use the centered-root and triangular elimination ideas of source62; their purpose here is to transfer the new deletion to additional degree slices.

Thirty-four integers, degree six. Make the triangular change $\tau_{\text{old}} = \tau_{\text{tail}} + E(c - \eta)$ and eliminate

$$P = \eta + \zeta + E(c - \eta), \quad N = q(L + 1), \quad X = Wq,$$
$$C_2 = c^2, \qquad F_2 = f^2, \qquad g = y^2 - V^2.$$

The first norm becomes exactly

$$\tau_{\text{tail}}^2 + (2\tau_{\text{tail}} - \eta - \zeta)E(c - \eta) - 1.$$

The leading quartic terms cancel, so every surviving row has degree at most three. Six coordinates and six defining rows disappear, leaving thirty-four integers and nineteen rows. The only row involving r is $I - (2r + 1)c^2$, so the coefficient of r^2c^4 in their sum of squares is exactly 4. The equation in `sextic34.json` has exact degree six.

Twenty-nine integers, degree eight. Put $k_0 = 1 - V + jc + hE$ and eliminate the eleven coordinates

$$L = q^2 - 1, \quad N = q^3, \quad X = Wq, \quad C_2 = c^2, \quad F_2 = f^2,$$
$$g = y^2 - V^2, \quad B_0 = I^2, \quad D = Wq + ac + \gamma(4a + 3),$$
$$\Delta = a^2 + 4a + 3, \quad \zeta = k_0 - \eta, \quad P = k_0 + E(c - \eta).$$

Every right-hand side uses retained coordinates only. The index-exposure row r_{17} supplies the monic definition of ζ, while r_7 supplies the restoration of P. Fourteen rows remain, all of degree at most four. The ordinary strong row

$$I^2 - (a^2 + 4a + 3)(f^2 - 1)$$

is the only row containing both a and f; its square contributes a^4f^4 with coefficient 1. Therefore the equation in `octic29.json` has exact degree eight. Both projections preserve the exact fixed tuple and inherit the same consistency equivalence.

2.8 A general principle and what the comparison means

Proposition 2.2 (Orientation after arithmetic recovery)**.** *Suppose a polynomial system, without a designated sign guard, has an arithmetic proof that every integer zero satisfies $X = \pm b^r$, where r lies in a specified set of integer indices. Let m be a fixed modulus such that the possible residues of b^r and $-b^r$ are disjoint. If affine restrictions on existing coordinates force X into the positive residue class, preserve a complete canonical witness family, and the deleted guard can be restored from the recovered positive power, then they replace that guard without introducing new unknowns.*

Proof. Disjointness excludes the negative branch of the recovered power. The original sign guard can then be restored if it follows from the recovered positive power, as $X = 2u^2$ does here for odd r. Canonical completeness supplies the converse existence implication. □

In this construction $b = 2$, $m = 3$, and the index is odd: the two residue sets are the singletons $\{2\}$ and $\{1\}$. The important requirement is the sign-uniform recovery theorem; simply dropping a square guard and imposing a residue would be unjustified without it.

At the inspected source pin the comparable variable counts are $41, 35, 30$ at degrees $4, 6, 8$. The new counts are $40, 34, 29$, with respectively $25, 19, 14$ residuals. Their exact expanded sums of squares, with the four fixed ports left as coefficient symbols, have $393, 614, 1238$ monomials when the direct quartic is selected; the two-affine-coordinate quartic used for projections has 614. The prior 41-variable quartic has 388. The affine restrictions enlarge these expanded expressions. Thus the gain is in unrestricted-integer variables and residual counts at fixed degree, not a decrease in every size measure. Native coefficient payload size is unchanged; no new bound for nine positive or ten unrestricted integer unknowns is asserted.

The standard-library builder and a separately authored verifier both check all literal affine substitutions, all twenty-five restoration identities for each projection, exact symbolic expansions, liveness, degrees, nonzero degree certificates, source hashes, and the actual coefficient payloads. Both pass in normal and optimized Python execution. These checks establish the finite algebraic claims. The unbounded Pell, native-compiler, and syntactic-refutation implications remain the stated mathematical proof and its named dependencies.

3 Transferring a strong Pell root into a transport gap

This section gives a polynomial in twelve strictly positive integer unknowns of exact degree 84, with a literal 103-operation arithmetic circuit. Its predecessor in package 61 has degree 118 and 104 operations. A thirteen-positive-variable companion has degree 30 and 94 operations, against package 61's degree 34 and 93 operations. These comparisons concern the actual circuits, count all live binary additions, subtractions and multiplications, and treat the complete fixed coefficient numerals as leaves.

The new twelve-variable construction removes the normalized strong Pell root through the first transport carrier. A positive upper approximation to that root makes both recovered root signs harmless. Strengthening an existing transport multiplier makes the inverse coordinate positive. Thus the construction does not require a new index congruence for the auxiliary Pell ordinate. The smaller thirteen-variable companion combines package 61's transport carrier with the adaptive modulus of package 65.

3.1 Fixed coefficients and the precise inherited boundary

Use the whole compact coefficient tuple of package 44:

$$\beta = 2^{590069} - 1, \qquad K_0, \qquad M_C, \qquad M_F.$$

The four integers are delivered as lossless payloads in `coefficients/`; M_F is already the shifted source mask. They are constants, not existential parameters. The fixed numerical binding and the unchanged source proof are recorded separately from the new reductions.

We use package 61's normalized positive-source interface. With positive coordinates $J, F, \alpha, z, f, h, i, T_{\mathrm{old}}, s, w, \tau, \eta, \zeta, y, \gamma_{\mathrm{old}}$, define

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, \\ X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, \\ c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, \\ \ell &= EkY, & I &= ic^2. \end{aligned}$$

$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J.$$

The six source equations are

$$\tau^2 - \ell(\ell + k) = 1, \qquad D^2 - \Delta c^2 = 1, \qquad D = X + ac + H\gamma_{\mathrm{old}}, \tag{18}$$

$$f^2 - \Delta I^2 = 1, \qquad \Delta(f^2 - 1)(V^2 - y^2) + y^2 = 1, \tag{19}$$

$$k - hE - R = 1, \qquad (K_0 + w)C + (q - F) - z(q - 1) = 1, \tag{20}$$

where $V = c(T_{\mathrm{old}} f - 1) - Rf^2$. The inherited compiler and K215 calibration identify existence of a positive solution with a finite syntactic ZFC refutation. This source interface, rather than a finite experiment, is the semantic premise used below.

Put $B = \beta + 1$. We use the source's canonical completeness promises

$$q = B^N, \quad J = 1 + B + \cdots + B^{N-1}, \quad N = 581197^e > 1, \quad R \equiv 3 \pmod 4, \quad X = 2^R.$$

The first Pell index is $n_0 = (R+1)/2 \geq 3$. A normalized strong solution can be chosen together with a positive auxiliary solution of the form

$$V = f^2(cT - R) - c, \qquad y = V + u, \qquad T, u > 0.$$

This is the normalized source and canonical construction of package 61's report, Sections 2 and 6, together with its positive auxiliary reconstruction lemma (Section 3, source label `lem:auxcomplete`); no stronger congruence for y is required here.

For arbitrary positive outer coordinates, once $C > 0$ has been proved, the authentic masks give

$$q \geq 16, \qquad 3q < R < q^4 - 2. \tag{21}$$

These bounds do not presuppose $q \mid X$.

3.2 A weighted transport carrier and the adaptive main carrier

Initially supply X, Y directly. Let $j, F, \alpha, f, i, T, X, Y, r, \eta, \zeta, u, \gamma$ be positive, put $J = 3j + 1$, and define the registers above. Let $h_* > 0$ be an integer polynomial register, and set

$$\begin{aligned} W &= 1 + (q-1)h_*, & M &= q^3(q-1), \\ B_t &= (qK_0 + X)C - qF, & Z &= M(\ell + k + r) + (q-1)Y - q^3 W B_t, \\ A_0 &= Z^2 - M^2(\ell(\ell+k)+1). \end{aligned} \tag{22}$$

The adaptive main carrier is

$$\begin{aligned} S &= 4qY, & U &= qS = 4q^2Y, & r_0 &= k - R - 1, \\ D_* &= U(X + ac) + H(U\gamma + SX - r_0), \\ A_1 &= D_*^2 - U^2(1 + \Delta c^2). \end{aligned} \tag{23}$$

Keep the other two residuals

$$\begin{aligned} A_s &= f^2 - \Delta I^2 - 1, \\ V &= f^2(cT - R) - c, \\ A_a &= (V+u)^2 - \Delta(f^2-1)u(2V+u) - 1. \end{aligned} \tag{24}$$

Weighted-carrier lemma. Every positive simultaneous zero of A_0, A_1, A_a, A_s, with any positive integer h_*, reconstructs a positive solution of (18)–(20). Conversely, canonical source witnesses lift whenever the chosen multiplier satisfies $h_* \geq k$.

Proof. If a rational number has integral square, it is an integer. Apply this to Z/M: $A_0 = 0$ supplies an integer τ with $\tau^2 = \ell(\ell+k)+1$. Since $M \mid Z$, reduction modulo q^3 gives $q^3 \mid Y$. Reduction modulo $q-1$ gives $(q-1) \mid B_t$, because $W \equiv q^3 \equiv 1 \pmod{q-1}$. Define the proof-only integers $s = Y/q^3 > 0$ and $t = B_t/(q-1)$. Dividing the carrier gives

$$\tau = \ell + k + r + s - Wt.$$

As $|\tau| < \ell + k$, we obtain $Wt > r + s > 0$, hence $t > 0$ and $B_t > 0$. If $C \leq 0$, then $B_t = (qK_0 + X)C - qF < 0$, a contradiction. Thus $C > 0$ and (21) is now available.

Write $A = a + 2$ and $P_0 = 2XY^2 + 1$. The main numerator is positive before either $q \mid X$ or rank recovery: since $r_0 < k$, $c > kY$ and $UaY > H$, we have $D_* > Uac - Hk > 0$. Thus $A_1 = 0$ gives a positive integer $D = D_*/U$ with $D^2 - \Delta c^2 = 1$. Pell classification supplies

$$D = \chi_A(p), \quad c = \psi_A(p), \qquad |\tau| = \chi_{P_0}(n_0), \quad k = 2\psi_{P_0}(n_0).$$

Here $p \geq 2$ and $n_0 \geq 2$. Indeed $n_0 = 1$ would give $k = 2$ and $c = 2Y + 1 < 2A = \psi_A(2)$, which is impossible. Consequently

$$k > 8XY^2, \qquad c > 8XY^3 \geq 8q^9 > 2R, \qquad n_0 < p, \qquad c > 2p.$$

The normalized strong residual implies $f = \chi_A(m)$, $I = \psi_A(m)$, $c \mid m$, and $m > 2p$; this is the normalized-rank lemma of package 61. Put $S_* = \Delta I$. The auxiliary residual gives

$$S_*^2 V^2 - (S_*^2 - 1)y^2 = 1, \qquad y = |V + u| > 0,$$

with $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. The signed auxiliary-index lemma, with $0 < p, R < c/2$, yields $p = R$. The rank lemmas used here do not require the deleted scale or index quotient, nor the sign of V.

It remains to extract the adaptive congruence without discarding a factor of three. Since 590069 is odd, $\beta \equiv 1 \pmod 3$ and $J = 3j + 1$ imply $q \equiv 2 \pmod 3$. Because $q^3 \mid Y$, $\gcd(q, H) = 1$. Moreover $H = 4Y(X + 1) + 3$ is odd and $\gcd(H, Y) = \gcd(3, Y)$. Hence

$$g := \gcd(H, U) = \gcd(H, S) \in \{1, 3\}.$$

Integrality of D_*/U gives $U/g \mid SX - r_0$, hence $S/g \mid r_0$. The first Pell recurrence gives

$$k \equiv 2n_0 \pmod{4XY^2}.$$

Since $S \mid 4XY^2$ and $S/g \geq 4q^4/3 > R + 1$, the inequalities $1 \leq n_0 < p = R$ force $2n_0 - R - 1 = 0$. Thus $r_0 = k - 2n_0$ is a positive multiple of $4XY^2$. Also $U \mid 4XY^2$, since $q^2 \mid Y$; therefore $U \mid r_0$. It follows from $U/g \mid SX$ that $q \mid gX$. Because g divides 3 and $3 \nmid q$, this gives $q \mid X$.

Now $w = X/q > 0$, $h = r_0/(XY) > 0$, and $\gamma_{\text{old}} = \gamma + w - r_0/U$ are integers. Since $D > (a+1)c$ and $c > X$, the identity $H\gamma_{\text{old}} = D - ac - X$ proves $\gamma_{\text{old}} > 0$. Since $B_t = q((K_0 + w)C - F) > 0$ and $(q - 1) \mid B_t$, the integer $z_{\text{new}} = ((K_0 + w)C - F)/(q - 1)$ is positive; set $z = z_{\text{new}} + 1$. Finally replace τ by $|\tau|$, put $T_{\text{old}} = fT$, and retain $y = |V + u|$. These are the required source coordinates.

For completeness, use canonical source witnesses. Their odd repunit exponent gives $J \equiv 1 \pmod 3$, so $j = (J - 1)/3 > 0$ is integral. The source transport coordinate has $z \geq 2$, hence $t = q(z - 1) \geq q$. Put

$$r = Wt + \tau - \ell - k - s.$$

Since $\tau > \ell$, $s < k$, and $h_* \geq k$, this gives $r > qW - 2k > 0$. It makes $A_0 = 0$. The canonical first index gives r_0 a positive multiple of $4XY^2$; take $\gamma = \gamma_{\text{old}} - w + r_0/U$. It is positive because $r_0/U \geq XY/q^2 = wY/q > w$. Its substitution makes $A_1 = 0$. The positive auxiliary reconstruction supplies T, u, and the two remaining residuals vanish. □

Taking $h_* = k$ defines the thirteen-variable companion

$$P_{13} = A_0^2 + A_1^2 + A_a^2 + A_s^2.$$

Its residual degrees are respectively $15, 13, 13, 14$, so its exact total degree is 30. The factor 4 in $S = 4qY$ is used in the proof: removing it without another argument loses the bound $S/g > R + 1$.

3.3 The twelve-variable polynomial

Supply exactly

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}.$$

Use the same common registers, but now define

$$\begin{aligned} D_f &= 1 + \Delta I^2, & L &= (a + 2)I, & h_* &= ikc, \\ W &= 1 + (q - 1)ikc, & Z_0 &= M(\ell + k + n) + (q - 1)Y - q^3 W B_t, & & \\ \overline{Z} &= Z_0 + ML, & D_\tau &= \ell(\ell + k) + 1. & & \end{aligned} \tag{25}$$

The first residual is the ordinary polynomial

$$N = \left(\overline{Z}^2 - M^2(D_f + D_\tau)\right)^2 - 4M^4 D_f D_\tau. \tag{26}$$

Retain A_1 from (23) and replace the auxiliary registers by

$$V = D_f(cT - R) - c, \qquad Q = \Delta(D_f - 1), \qquad A_a = (V + u)^2 - Qu(2V + u) - 1. \tag{27}$$

The recommended equation is

$$\boxed{P_{12} = N^2 + A_1^2 + A_a^2 = 0.} \tag{28}$$

Every named register is a polynomial abbreviation. There are twelve, not twelve plus the number of registers, quantified unknowns.

Root-transfer theorem. The polynomial (28) has a strictly positive zero if and only if the inherited source has a strictly positive zero. Consequently its lack of positive zeros is equivalent in ZFC to Con(ZFC), subject exactly to the inherited compiler/calibration interface above.

Proof. The inequalities needed to recover roots hold on every positive input, before any residual is assumed zero. Since $\ell < ac$ and $k < c$,

$$\ell(\ell + k) < a(a + 1)c^2 < \Delta i^2 c^4,$$

and therefore $D_f > D_\tau > 0$. Also

$$L^2 - D_f = I^2 - 1 > 0,$$

because $c \geq 3$ and $I = ic^2 > 1$.

At a zero of N, $\overline{Z} = 0$ is impossible: it would imply $M^4(D_f - D_\tau)^2 = 0$. The identity

$$N = \left(\overline{Z}^2 + M^2(D_f - D_\tau)\right)^2 - 4M^2\overline{Z}^2 D_f$$

then shows that the rational number

$$f_* = \frac{\overline{Z}^2 + M^2(D_f - D_\tau)}{2M\overline{Z}}$$

has square D_f. It is a nonzero integer by rational-square integrality. Furthermore

$$\left(\frac{\overline{Z} - Mf_*}{M}\right)^2 = D_\tau,$$

so this second rational root is also an integer. Set $f = |f_*|$ and

$$r = n + L - f_*.$$

The strict inequality $L > |f_*|$ gives $r > 0$ for both signs of f_*. The first carrier becomes $Z = \overline{Z} - Mf_*$, so $A_0 = 0$. The strong residual is zero because $f^2 = D_f$. All other registers involving the strong root use its square only; thus the supplied $A_1 = A_a = 0$ give the remaining weighted-carrier equations. The lemma reconstructs a positive source solution.

Conversely, use canonical source witnesses and their positive strong root f. Choose T, u by positive auxiliary reconstruction, and r, γ by the weighted-carrier lemma with $h_* = ikc$. This multiplier is at least k. The upper approximation gap satisfies

$$0 < L - f = \frac{I^2 - 1}{L + f} < \frac{I}{a} = \frac{ic^2}{a} < ikc,$$

because $c = kY + \eta < k(Y+1) \le ka$ for $X, Y \ge 1$. The transport construction gives

$$r > qW - 2k > ikc.$$

The last inequality follows directly from $q \ge 16$ and $ic \ge 3$. Hence $n = r - L + f$ is a strictly positive integer. Its substitution gives $\overline{Z} = M(\tau + f)$, and the two source squares imply $N = 0$. The other two residuals already vanish. This proves completeness, including positivity of the new coordinate. □

The theorem proves an equivalence and does not decide either side. The norm is used with its zero-denominator exclusion and both signed-root branches; an abstract appeal to independent square classes would not supply these facts.

3.4 Exact degree, arithmetic cost, and finite certificates

Write

$$q_0 = 3\beta j, \quad k_0 = \eta + \zeta, \quad B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

For the twelve-variable equation,

$$\deg M = 4, \quad \deg L = 7, \quad \deg D_f = 14, \quad \deg D_\tau = 8, \quad \deg Z_0 = 10.$$

The cancellation-resolved middle expression in (26) is

$$\overline{Z}^2 - M^2(D_f + D_\tau) = Z_0^2 + 2MLZ_0 - M^2(1 - I^2 + D_\tau). \tag{29}$$

It has degree 21; only its middle summand reaches that degree. The subtracted norm tail $4M^4 D_f D_\tau$ has degree 38. Thus N has exact degree 42, with leading homogeneous form

$$\mathrm{in}_{42}(N) = 4q_0^{16} i^4 k_0^8 X^2 Y^8 B_{t,0}^2.$$

The other residual degrees are 13 and 37, respectively. Consequently

$$\mathrm{in}_{84}(P_{12}) = 16 q_0^{32} i^8 k_0^{16} X^4 Y^{16} B_{t,0}^4. \tag{30}$$

In particular the coefficient of $j^{32} i^8 \eta^{16} X^8 Y^{16} \alpha^4$ is exactly $16(3\beta)^{32} > 0$. Choosing this minimum j exponent forces the $-X\alpha$ term from each copy of $B_{t,0}$, so the other fixed ports cannot cancel it. This proves exact degree after the actual coefficient specialization.

For P_{13}, the unique highest residual degree is 15 and

$$\mathrm{in}_{15}(A_0) = -2q_0^8 k_0^2 XY^2 B_{t,0}, \qquad \mathrm{in}_{30}(P_{13}) = 4q_0^{16} k_0^4 X^2 Y^4 B_{t,0}^2.$$

Its coefficient of $j^{16} \eta^4 X^4 Y^4 \alpha^2$ is $4(3\beta)^{16} > 0$.

The delivered twelve-variable evaluator uses 60 multiplications, 27 additions and 16 subtractions, totaling 103 live operations. The three-equation version uses 98 operations and residual degrees $42, 13, 37$. The thirteen-variable evaluator uses 94 operations, or 87 for its four-equation system. The JSON graphs contain no divisions or variable exponents.

The verifier checks literal graph liveness, every fixed-payload digest, byte-identical circuit regeneration, exact prefix identities, independent cut-ring norm and finalizer identities, and the nonzero leading forms above. These are finite certificates of algebra and data. The unbounded existence equivalence follows from the written proofs and the inherited source theorem, not from bounded tests. No new end-to-end Lean/Rocq certificate or exported object-language ZFC derivation is claimed for these literal polynomials.

4 Three-root transport: eleven positive variables, degree 325

There is a division-free polynomial in eleven strictly positive integer unknowns of exact total degree 325, evaluated by 113 binary arithmetic operations, whose lack of positive solutions is equivalent in ZFC to Con(ZFC). At the pinned repository state the eleven-positive equation in package 52 has degree 954 and the same 113-operation cost. The construction below removes 629 degrees while retaining that operation count. Its two residuals have degrees 312 and 13 and together require 110 operations.

The new ingredient is a particular combination of the normalized positive auxiliary reconstruction, the weighted first carrier, and an elementary denominator argument. Conjugate norms, positive majorants, radical separation, quotient absorption and unit-product finalizers already occur in package 52 and its predecessors; those general methods are not claimed as new. In package 52 the auxiliary radicand depends on the signed strong Pell root and the main Pell root is eliminated inside the same norm. Here the normalized auxiliary expression depends only on its square. The first, strong and auxiliary roots therefore enter a direct octic sum norm, while the inexpensive adaptive main equation remains separate.

4.1 Fixed data and the literal equation

Use precisely the coherent tuple (β, K_0, M_C, M_F) of Section 3, with $\beta = 2^{590069} - 1$ and the already shifted M_F. These are fixed integer numerals, not parameters. Their payloads and source hashes are supplied again in this component. The authentic mask inequalities used below are

$$0 < M_C < \beta, \qquad 0 < M_F < 2\beta, \qquad K_0 > 0.$$

The eleven unknowns are exactly

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma \in \mathbb{Z}_{>0}.$$

Define polynomial registers

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1,\\
\ell &= EkY, & M &= q^3(q-1), & B_t &= (qK_0 + X)C - qF,\\
D_\tau &= \ell(\ell + k) + 1.
\end{aligned}$$

Set

$$\begin{aligned}
R &= \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J, \qquad (31)\\
\widehat{R} &= q^3(q + F + \alpha),
\end{aligned}$$

and then

$$\begin{aligned}
I &= iq(q-1)c^2, & D_f &= 1 + \Delta I^2, & Q &= \Delta(D_f - 1), \qquad (32)\\
\delta &= Q - 1, & V &= D_f(cT - R) - c, & D_y^{\mathrm{num}} &= QV^2 - 1,\\
\overline{V} &= D_f(cT + \widehat{R}), & L &= 3\overline{V}, & W &= 1 + (q-1)L. \qquad (33)
\end{aligned}$$

The first transport register is

$$\overline{Z} = M(\ell + k + n + L) + (q-1)Y - q^3 W B_t. \tag{34}$$

The adaptive main carrier is unchanged:

$$\begin{aligned} &r_0 = k - R - 1, && S = 4qY, && U = qS, \\ &D_* = U(X + ac) + H(U\gamma + SX - r_0), \\ &A_1 = D_*^2 - U^2(1 + \Delta c^2). \end{aligned} \tag{35}$$

The following schedule evaluates the octic norm using only integer polynomial operations:

$$\begin{aligned} &\mathcal{A} = M^2 D_\tau, && \mathcal{B} = M^2 D_f, && \mathcal{C} = M^2 D_y^{\mathrm{num}}, \\ &T_n = \delta^2 \mathcal{A}\mathcal{B}, && R_n = \delta \mathcal{C}, && J_n = \delta(\overline{Z}^2 - \mathcal{A} - \mathcal{B}) - \mathcal{C}, \\ &N_* = J_n^2 - 4\big(T_n + R_n(\mathcal{A} + \mathcal{B})\big), \\ &N_8 = N_*^2 - 64 T_n R_n \overline{Z}^2. \end{aligned} \tag{36}$$

The equation is

$$\boxed{P_{11} = N_8 + A_1 + N_8 A_1 = 0.} \tag{37}$$

All the displayed registers are abbreviations, with no additional quantified witnesses. The supplied JSON is a literal binary $+, -, \times$ circuit; even multiplication by a small integer is charged. The rational quantity

$$D_y = \frac{D_y^{\mathrm{num}}}{\delta}$$

is used only in the proof and is never evaluated by the polynomial circuit.

4.2 An elementary rational-root extraction lemma

Lemma 4.1. *Let d_1, d_2, d_3 be distinct positive rational numbers. Suppose signed real roots $x_s^2 = d_s$ have a nonzero rational sum $t = x_1 + x_2 + x_3$. Then all three x_s are rational.*

Proof. Put

$$u = \frac{t^2 - d_1 - d_2 - d_3}{2}, \qquad v = \frac{u^2 - d_1 d_2 - d_1 d_3 - d_2 d_3}{2t}.$$

These are rational. Expansion gives $u = x_1x_2 + x_1x_3 + x_2x_3$ and $v = x_1x_2x_3$. Consequently

$$d_1 t + v = x_1(x_1 + x_2)(x_1 + x_3), \qquad x_1 = \frac{d_1(d_1 + u)}{d_1 t + v}.$$

The denominator is nonzero. Indeed, $x_1 \neq 0$, while either $x_1 = -x_2$ or $x_1 = -x_3$ would contradict distinctness of the radicands. The two cyclic formulas recover x_2, x_3 as rational numbers as well. This proof accommodates every coincidence or dependence among square classes; it assumes no independent specialized Galois automorphisms. □

4.3 Unconditional radical separation and the positive guard

All bounds in this subsection hold at every positive input tuple, before assuming any residual vanishes. In particular they do not assume $C > 0$, $q^3 \mid Y$, or a recovered Pell index. We have

$$q \geq 16, \quad k \geq 2, \quad c \geq 3, \quad a \geq 2, \quad \Delta \geq 15, \quad I \geq c^2, \quad \ell < ac, \quad k < c.$$

Thus

$$\ell(\ell + k) < a(a + 1)c^2 < \Delta I^2, \qquad 1 < D_\tau < D_f < Q.$$

Write $t = \sqrt{D_\tau}$ and $f = \sqrt{D_f}$ for positive real roots. Then

$$t < \ell + k < (a + 1)c, \qquad f > (a + 1)I \geq (a + 1)c^2 \geq 3c^2. \tag{38}$$

The integer $b = cT - R$ determines two cases. If $b = 0$, then $V = -c$. If $b \neq 0$, then $|V| \geq D_f - c > 1$. Hence $|V| > 1$ in both cases and

$$D_y = V^2 + \frac{V^2 - 1}{Q - 1} > V^2 > 1. \tag{39}$$

The radicands D_τ, D_f, D_y are pairwise distinct. For if $D_y = D$ with $D \in \{D_\tau, D_f\}$, then

$$Q(D - V^2) = D - 1,$$

which is impossible because $0 < D - 1 < Q$.

Let $y = \sqrt{D_y} > 0$. If $b = 0$, then $Q > c^2$ implies $D_y < c^2 + 1$ and $y < c + 1$, whereas (38) gives

$$f - t > (a + 1)c(c - 1) > c + 1.$$

Thus $f > t + y$. If $b \neq 0$, the bound $f > 3c^2 > c + 2$ implies $f^2 - c > 2f$, so

$$y > |V| \geq D_f - c > 2f > f + t.$$

In either case one root exceeds the sum of the other two. Therefore no signed sum $\pm t \pm f \pm y$ is zero.

It remains to check that the polynomial L bounds the two roots to be transferred out of the positive coordinate. The authentic masks give

$$(M_C + qM_F)J < (q - 1)(1 + 2q) < 2q^2.$$

The triangle inequality applied to (31) then gives

$$|R| < q^4 + Fq^3 + (\alpha + 2)q^2 < q^3(q + F + \alpha) = \widehat{R},$$

because $\alpha(q - 1) > 2$. Both sides are integers, and $D_f > c$; hence

$$|V| \leq D_f(cT + |R|) + c < D_f(cT + \widehat{R}) = \overline{V}.$$

Also $\overline{V} > D_f > f$ and, by (39), $y < 2|V| < 2\overline{V}$. We obtain the uniform bounds

$$L = 3\overline{V} > f + y, \qquad L > c > k. \tag{40}$$

4.4 What the cleared norm forces

The universal octic identity is

$$\begin{aligned} F_8(x;a,b,c) &:= \left((x^2-a-b-c)^2-4(ab+ac+bc)\right)^2-64abcx^2 \\ &= \prod_{\epsilon_1,\epsilon_2,\epsilon_3\in\{-1,1\}} (x-\epsilon_1\sqrt{a}-\epsilon_2\sqrt{b}-\epsilon_3\sqrt{c}). \end{aligned} \tag{41}$$

This is a polynomial identity, verified by multiplying the eight formal conjugates. The center in (36) also equals

$$\delta^2\left((\overline{Z}^2-\mathcal{A}-\mathcal{B})^2-4\mathcal{A}\mathcal{B}\right)-2\delta\mathcal{C}(\overline{Z}^2+\mathcal{A}+\mathcal{B})+\mathcal{C}^2.$$

Consequently

$$N_8=\delta^4F_8(\overline{Z};M^2D_\tau,M^2D_f,M^2D_y). \tag{42}$$

All three specialized radicands are positive, and $M,\delta>0$. If $N_8=0$, one of its eight real factors vanishes. There are signed roots satisfying

$$\frac{\overline{Z}}{M}=\tau_*+f_*+y_*, \qquad \tau_*^2=D_\tau, \quad f_*^2=D_f, \quad y_*^2=D_y. \tag{43}$$

The preceding separation proves that this rational sum is nonzero and its three radicands are distinct. Lemma 4.1 makes all three signed roots rational. Since D_τ, D_f are integers, rational-square integrality yields $\tau_*, f_*\in\mathbb{Z}$.

The auxiliary root becomes integral for a different, elementary reason. Let d be its positive reduced denominator. Equation (43) gives $d\mid M$, and $y_*^2=D_y^{\mathrm{num}}/\delta$ gives $d^2\mid\delta$. Every prime divisor of $M=q^3(q-1)$ divides $q(q-1)$ and therefore divides $I=iq(q-1)c^2$ and $Q=\Delta^2I^2$. It cannot divide $\delta=Q-1$. Thus

$$\gcd(M,\delta)=1, \qquad d=1,$$

and y_* is an integer. This uses only the prime support of M; no claim that $M\mid Q$ is needed.

4.5 Soundness and completeness

Theorem 4.2. *The equation (37) has a strictly positive solution if and only if the inherited normalized compiler has a strictly positive solution. Hence, using the same compiler and refutation calibration, ZFC proves that (37) has no positive solution if and only if* Con(ZFC).

Proof. First, the single equation is equivalent to $N_8=A_1=0$. Indeed,

$$P_{11}=(1+N_8)(1+A_1)-1.$$

Both residuals are squares modulo 4: this follows directly from (36) and from $4\mid U$ in (35). At a zero of P_{11}, the two integer factors are therefore both 1; the alternative of both being -1 would force a residual to be $-2\equiv 2 \pmod 4$. The converse is immediate.

For soundness, recover the signed integral roots in (43), and set

$$i_{\mathrm{old}}=iq(q-1), \quad f_{\mathrm{old}}=|f_*|, \quad y_{\mathrm{old}}=|y_*|, \quad u=y_{\mathrm{old}}-V, \quad r=n+L-f_*-y_*.$$

Equation (39) gives $y_{\mathrm{old}}>|V|$, so $u>0$. The uniform bound (40) gives $r>0$ for every sign choice. The strong equation is

$$f_{\mathrm{old}}^2-\Delta(i_{\mathrm{old}}c^2)^2=1.$$

The auxiliary root equation is exactly

$$y_{\mathrm{old}}^2 - Q(y_{\mathrm{old}}^2 - V^2) = 1,$$

which is the normalized auxiliary residual with $y_{\mathrm{old}} = V + u$. Finally the first transport register with weight $h_* = L$ becomes

$$Z = M(\ell + k + r) + (q-1)Y - q^3 W B_t = \overline{Z} - M(f_* + y_*) = M\tau_*.$$

It satisfies $Z^2 = M^2 D_\tau$. The weighted transport and adaptive main carrier theorem from Section 3 now applies, with all its witnesses positive, and recovers a normalized compiler solution. The signed roots must be retained in the definition of r; absolute values are taken only in the strong and auxiliary witnesses.

For completeness, start with a canonical normalized compiler solution. Keep its outer quantities, including q, X, Y, k, c, a, R and its first positive Pell root τ. The canonical facts needed from the weighted transport theorem are

$$R \equiv 3 \pmod 4, \quad c = \psi_{a+2}(R), \quad t_0 = \frac{B_t}{q-1} = q(z_{\mathrm{source}} - 1) \geq q, \quad s = \frac{Y}{q^3} < k, \quad \tau > \ell.$$

We must choose the strong Pell solution with the additional divisibility $q(q-1)c^2 \mid I$; this is not an assumption about a preselected source witness. Put $m_0 = q(q-1)c^2$ and consider

$$U_0 = \begin{pmatrix} a+2 & \Delta \\ 1 & a+2 \end{pmatrix}.$$

Its determinant is 1, so its reduction belongs to the finite group $\mathrm{SL}_2(\mathbb{Z}/m_0\mathbb{Z})$. A positive power is the identity. Since the top-left and lower-left entries of U_0^v are $\chi_{a+2}(v)$ and $\psi_{a+2}(v)$, a positive multiple of that order gives positive integers f, I with

$$f^2 - \Delta I^2 = 1, \qquad m_0 \mid I.$$

Set $i = I/m_0 > 0$. The positive auxiliary reconstruction lemma in the pinned package 61 report (its label `lem:auxcomplete`) applies to every such strong solution: its hypotheses are $R \equiv 3 \pmod 4$, $c = \psi_{a+2}(R)$ and $c \mid I$. It supplies positive T, u and $y = V + u > 0$ satisfying the normalized auxiliary equation. Thus the stronger divisibility preserves the required reconstruction promises.

Only after choosing these roots and T do we compute $\widehat{R}, \overline{V}, L, W$. Choose

$$r = Wt_0 + \tau - \ell - k - s.$$

Then

$$r > qW - 2k = q + q(q-1)L - 2k > L,$$

using $q \geq 16$ and $L > k$. Consequently

$$n = r - L + f + y > 0.$$

It follows that $\overline{Z} = M(\tau + f + y)$, so the norm vanishes. The unchanged adaptive main construction supplies positive γ with $A_1 = 0$. Its canonical repunit choice also gives $J = 3j + 1$ with $j > 0$, as in Section 3. This completes the lift. The order of choices was the strong pair, the auxiliary pair, the majorant, and then the transport coordinate; there is no circular choice of witnesses or unproved enlargement of a special index. □

4.6 Exact degree, circuit cost and verification boundary

The ordinary total degrees in the eleven supplied unknowns are

register	I	D_f	Q	V	L	W	$\overline{Z}$	D_y^{num}	$\mathcal{C}$
degree	7	18	22	22	22	23	28	66	74

and $\deg \mathcal{A} = 16$, $\deg \mathcal{B} = 26$. In the expanded expression for N_*, the term $\delta^2 \overline{Z}^4$ has degree 156, while the other two groups have degrees at most 152 and 148. The norm tail has degree 238, below $2 \cdot 156$. Thus the top part of N_8 is $Q_0^4 \overline{Z}_0^8$, of degree 312.

For an explicit certificate, set

$$q_0 = 3\beta j, \qquad k_0 = \eta + \zeta, \qquad B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F.$$

The relevant leading forms are

$$\begin{aligned} Q_0 &= i^2 q_0^4 k_0^4 X^4 Y^8, \\ \overline{Z}_0 &= -3 i^2 q_0^{11} k_0^4 X^2 Y^6 (q_0 + F + \alpha) B_{t,0}, \\ (A_1)_{13} &= 32 q_0^2 k_0 X^2 Y^4 \big(4 q_0^2 Y \gamma + q_0^3 (q_0 - F)\big). \end{aligned}$$

The apparent degree-14 terms of A_1 cancel; the last expression is verified by exact sparse polynomial expansion, not by gate-degree propagation. Therefore the degree-325 part of P_{11} is

$$2^5 3^8 \, i^{24} q_0^{106} k_0^{49} X^{34} Y^{84} (q_0 + F + \alpha)^8 B_{t,0}^8 \big(4 q_0^2 Y \gamma + q_0^3 (q_0 - F)\big).$$

Its coefficient of

$$j^{108} i^{24} \eta^{49} X^{42} Y^{85} \alpha^{16} \gamma \tag{44}$$

is exactly

$$2^7 3^8 (3\beta)^{108} > 0.$$

Indeed the γ term fixes the final factor, and minimum j degree reduces the two remaining parenthesized powers to $X^8 (F + \alpha)^{16}$, whose α^{16} coefficient is 1. This certifies degree 325 on the actual fixed coefficient slice; no specialization of K_0, M_C, M_F can cancel the selected coefficient.

The literal circuit has 67 multiplications, 29 additions and 17 subtractions, for 113 operations. Removing the three-operation finalizer leaves 110 operations for the two-residual system. The normal and optimized-Python replay receipts check byte-identical circuit regeneration, input and gate liveness, every polynomial register contract, the formal octic identity, denominator clearing, rational-extraction identities, the main-Pell cancellation, the exact degree certificate, and the complete fixed payloads and retained source hashes. An independent audit uses a separate sparse-ring implementation and does not import the construction builder.

These checks certify finite algebra and data integrity. The existence reduction is the proof above together with the preceding weighted carrier theorem; the unbounded compiler and K215 refutation calibration remain inherited mathematical dependencies. No end-to-end proof-assistant certificate, global minimality theorem, or proof of consistency is claimed.

5 Logical status, exact data, and reproducibility

5.1 The proof chain and its limits

There are three logically different kinds of evidence in this release. First are ordinary unbounded mathematical proofs: the signed power recovery and residue orientation, the root-transfer lemma, adaptive congruence recovery, and soundness and completeness of the new equations. Second are inherited mathematical theorems: the Pell rank/auxiliary interfaces, the native finite-window compiler, its fixed coefficient adapters, and the K215 equivalence with the specified syntactic refutation predicate. Third are finite checks of literal polynomials and data. None of the latter replace the first two.

For Q_{40}^{dir} the proof restores the deleted square coordinate and hence a zero of the parent Q_{41}. This deliberately keeps the native semantic interface unchanged. The companion eliminations are exact integer-polynomial bijections of zero sets. The affine restrictions used for completeness have a weaker and sufficient property: the canonical witness family lies in their image. We do not claim that every parent zero lies in that image.

For P_{12} the norm recovers a signed integer root f_* and a separate integer first root. The gap uses $r = n + L - f_*$; the positive strong coordinate is $|f_*|$. These two roles must not be conflated. The normalized source interface is then reconstructed with positive scale, transport, and main-root quotients. No division is used in the literal equation, and every division appearing in the proof is justified before its result is used as a witness.

For P_{11} the analogous recovery uses three signed roots. Pairwise distinct rational radicands and a nonzero rational sum force all three roots to be rational. The first two are integral because their squares are integers. For the third root, its reduced denominator divides $M = q^3(q-1)$ while the square of that denominator divides $Q - 1$; the engineered condition $\gcd(M, Q-1) = 1$ removes it. The proof restores positive gap and auxiliary coordinates for every recovered sign choice. Canonical completeness follows from a finite Pell-matrix period, so the stronger divisibility is proved rather than presumed.

The active repository separately reports checked Lean consistency equations and ongoing formalization of smaller arithmetic circuits. Its README explicitly distinguishes a Lean theorem about a syntactic predicate from an exported proof inside its object-language ZFC calculus [2]. This release does not run a Lean or Rocq kernel, does not claim a new complete formal certificate, and does not transfer the formal status of an existing equation to a different literal polynomial. The asserted equivalences are ordinary mathematical proofs formalizable in ZFC with the named inherited results. The independent reviews here are internal AI-assisted reviews, not external peer review.

5.2 Complete coefficient bindings

Two different complete four-port tuples are used. They must not be mixed. The integer family uses the source-34 mixed-parity unrounded tuple with $\beta = 2^{1064921} - 1$. The positive family uses the compact source-44 tuple with $\beta = 2^{590069} - 1$. The tuple members are β, K_0, M_C, M_F; M_F already includes the source's prescribed shift. Reapplying that shift changes the equation.

The supplied compressed integer payloads have exact, documented byte interpretations: after gzip decompression, bytes $b_0, \ldots, b_{m-1}$ represent $\sum_{j=0}^{m-1} b_j 256^j$. Thus the four "ports" in the symbolic polynomial files are fixed ordinary integer coefficients, not free parameters or uncomputed choices.

Their recipes, byte hashes, and source pins are included. No numeral is obtained by searching for a contradiction or by deciding a consistency statement.

The four raw native payloads total 532,479 bytes for the integer family and 295,388 bytes for the positive family. This is storage for the native inputs to the displayed polynomial recipes, not the size of a fully expanded specialized polynomial. The new affine restrictions change small integer multipliers in the expanded equations. Keeping the native tuple unchanged therefore does not mean that every expanded coefficient is unchanged.

5.3 What the verification programs establish

The root verification command authenticates the release manifest, makes an isolated copy, and runs the component verification there. It compares regenerated literal equation artifacts with the delivered bytes. Each component uses the Python standard library, with checks that remain active under `python3 -O`. The checks cover:

1. exact affine substitutions from the pinned parent and deletion of precisely the intended coordinate and residual;
2. exact polynomial restoration maps for both integer companions;
3. counts, complete coordinate use, residual degrees, and nonzero monomial certificates of the claimed total degrees;
4. literal positive arithmetic DAGs, their live operation counts, cancellation-resolved degree identities, and fixed-numeral binding;
5. source-file identities, coefficient payload integrity, and the included finite native coefficient reconstruction.

The integer audit also expands the sums of squares with the four fixed ports temporarily treated as symbolic coefficients. Its special monomial certificates are independent of their large values. The positive degree audit gives explicit nonzero leading coefficients after specialization to the actual positive β. These lower bounds are necessary: a degree propagated through a circuit is only an upper bound until possible cancellation is addressed.

No verification command runs an unbounded refutation search, supplies a zero of the consistency polynomial, or establishes unsolvability. The polynomial coefficients are completely materialized, while the existence equivalence is a theorem about all integer assignments.

5.4 Using and integrating the archive

The root directory contains this complete article as `report.tex` and `report.pdf`, the component files under `integer/`, `positive/`, and `eleven/`, comparison provenance under `baseline/`, checks and receipts, licenses, and a root verification driver. Paths mentioned inside component proofs are relative to that component. The cover and `README.md` identify the three main literal packets. All residual systems in the integer JSON files define one equation by taking their sum of squares. The twelve- and thirteen-positive JSON files include their sum-of-squares output nodes; the eleven-positive file uses the proved unit-product finalizer.

Run `python3 verify.py` from the extracted root, or add `-optimized` to exercise the checks with Python optimization enabled. Run `python3 build_report.py` to compile the article twice with `pdflatex`. The mathematical tests need no network access; the PDF build needs an ordinary TeX installation. PDF metadata can vary on recompilation, so the report rebuild is separate from exact equation regeneration.

The directory is prepared as an additive contribution. It can be placed beside the earlier numbered packages at intake without changing their delivered content or the active Lean equation. The source snapshot is fixed even if the repository subsequently advances.

6 Further research questions

Can more sign witnesses be replaced by residues? The essential new theorem recovers a finite set of possible algebraic values before using a congruence to select one. A next target is to classify which sign constraints in other low-degree Diophantine compilers have this property. A proposed deletion must prove the full signed version of the earlier argument and preserve a canonical completeness family. Checking that the intended witnesses satisfy a residue is insufficient to exclude unintended roots.

Can the common-sign guard be removed? The quartic still uses three coordinates to force $\eta\zeta > 0$. They supply the sharp interval $K(Z-1) < |c| < K(Z+1)$ needed for the rank comparison. Replacing this guard would save more variables than the single square witness removed here, but the interval cannot simply be assumed after removing it. A useful intermediate theorem would recover a suitable ratio interval from retained congruences and norms alone.

Can a fourth root be transferred equally cheaply? The eleven-variable construction resolves the auxiliary integrality obstacle through coprimality. Eliminating the separately retained main Pell carrier could improve another degree slice, but the three-root rationality lemma does not extend by the same hypotheses alone. For example,

$$10 + \sqrt{2} + \sqrt{8} - \sqrt{18} = 10$$

has pairwise distinct radicands and a root larger than the sum of the other three, yet three roots are irrational. A four-root construction needs control of proper signed subsums or another extraction mechanism, as well as a new integrality and positive-completeness argument. The larger nested norms in the predecessors provide a concrete comparison.

Can the transport multiplier be made less expensive? In the twelve-variable equation, $W = 1 + (q-1)ikc$ pays for the positive inverse gap. The degree-84 leading form shows precisely where this cost enters. A smaller sufficient polynomial majorant for $L - f$, compatible with the same congruence, could lower degree further. Such an estimate must be uniform in the cofinal strong-Pell witness, not just true for a few small indices.

Which reductions are best after fixing a description language? Native tuple bytes, expanded monomials, circuit operations, and proof complexity are different objectives. A useful next comparison would fix a concrete encoding of polynomial arithmetic DAGs and coefficients,

then compare total description length. The present release supplies enough literal data for that comparison; it does not infer it from variable count alone.

Can a direct proof-language compiler become competitive? Relational NBG and other finitely presented equiconsistent systems remain plausible alternatives to machine tableaux. The missing deliverable is a complete source predicate, a proved bridge to the specific syntactic consistency statement, and a measured explicit polynomial after all bounded-sequence and exponentiation devices are compiled. A smaller informal interpreter is not by itself a smaller Diophantine equation.

What should be formalized next? The signed power-recovery lemma is a focused target: its inputs are elementary Pell classifications, explicit magnitude bounds, and the already used rank interfaces. The two-root transfer lemma is an even smaller independent target. After those lemmas, the affine substitutions, restoration maps, and literal DAG identities can be connected to the repository's formal arithmetic syntax. Completing the native semantic and object-language internalization boundaries remains a separate task.

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